LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 10: The Game Loop and Collisions

Module 4 | 45 minutes | Game loop and collision detection | Grades 6–8 · ages 11–14

Objective: Build a game loop that moves objects, recycles them when they leave the screen, and detects collisions to change lives.

Success criteria

- I can build a game loop that moves and recycles objects.
- I can detect collisions and update lives.
- I can stop one hit counting many times.

Key vocabulary

- **game loop**: The loop that updates the game every frame.
- **collision**: Two objects overlapping.
- **recycle**: Reuse an object by moving it back to the start.
- **state**: Values that describe the game right now (lives, score).

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Dodgeball walk (5 min): one learner walks a line while soft balls roll; count “collisions”.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Rock dodger: learners follow the steps.
14 min	**Perform**	Code Lab: recycle the rock, Code Lab: lose a life
9 min	**Probe**	Debug the collision, activity
4 min	**Wrap-up**	Exit ticket: What should happen to the rock right after it hits the player?

Unplugged warm-up

Dodgeball walk (5 min): one learner walks a line while soft balls roll; count “collisions”.

Teaching the idea

A game loop updates, detects collisions and responds each frame. Object recycling avoids creating new sprites. Collision handling must remove the overlap (reset position) to avoid repeated triggers. State variables (lives) reset on start.

Analogy: A juggler: every moment, check each ball, catch or throw — over and over.

Live demo: Show the multi-hit bug, then fix by resetting the rock.

Questions to ask

Ask	Listen for
Why recycle the rock?	Endless objects from one sprite.
Why reset the rock after a hit?	So one hit counts once.
What is the game state?	Lives, score, positions.

Differentiation

Support
Provide the falling loop; learners add recycle and collision.

Stretch
Add a second rock with a different speed.

Common misconceptions

Misconception	How to fix it
touching is checked once	It is checked every loop.
More sprites are needed for more rocks	Recycling reuses one sprite.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What should happen to the rock right after it hits the player?

Homework
Watch a mobile game for 1 minute and list what its game loop checks.

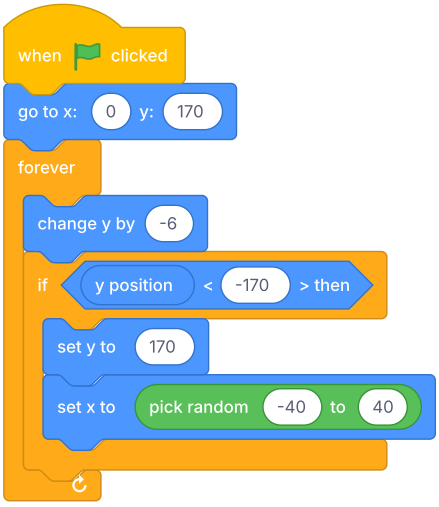
Answer key — Kitabash activities

1. Code Lab: recycle the rock (perform · code lab)

Goals checked: Falls forever · Comes back to the top · Random position · Keeps going

One solution:

Rock

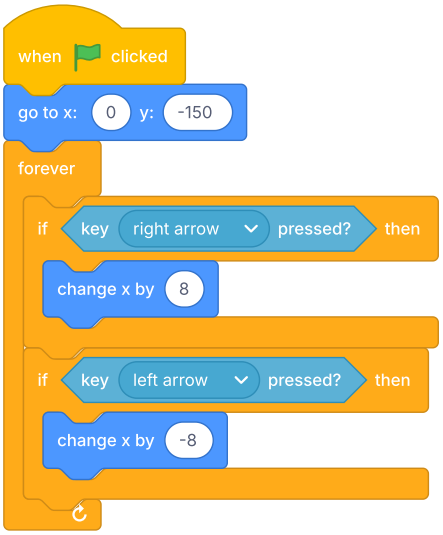


2. Code Lab: lose a life (perform · code lab)

Goals checked: Lives start at 3 · A hit costs a life · One life per hit

One solution:

Player



Rock



3. Debug the collision (probe · multiple choice)

Answer: Move the rock back to the top after the hit

4. Explain your thinking (probe · free response)

Look for: loop, frame, if. **Example:** Every frame the rock moves down and the player checks the arrow keys. Only when the rock is below the bottom does it jump back to the top. Only when it touches the player do we lose a life and reset the rock.

Answer key — worksheet

1. **Key words:** game loop = The loop that updates the game every frame. · collision = Two objects overlapping. · recycle = Reuse an object by moving it back to the start. · state = Values that describe the game right now (lives, score).
2. **Plan your code:** Goals: Falls forever; Comes back to the top; Random position; Keeps going.
3. **Plan your code:** Goals: Lives start at 3; A hit costs a life; One life per hit.
4. **Debug the collision:** A — The rock keeps touching for several frames; moving it away makes it count once.
5. **Explain your thinking:** Every frame the rock moves down and the player checks the arrow keys. Only when the rock is below the bottom does it jump back to the top. Only when it touches the player do we lose a life and reset the rock.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.